Combining Relevant Data in Excel

Objectives

Data Analytics often requires data from multiple sources. Keeping this data in sync with its various sources can be difficult. In this activity, you will link data in one worksheet to data contained in multiple other worksheets. You will use the Workbook Links feature of Excel to create external links and ensure that data in the multiple worksheets remains synchronized.

External references between Excel workbooks are especially useful when it's not practical to keep large worksheet models together in the same workbook. You can use these references to:

Consolidate data from several workbooks - Workbooks that contain data from several years, departments, or products can be linked. You can perform summary or other functions on the data to create reports or graphs.

Create different subsets of your data - You can use external references to create reports that contain only relevant data, not the entire contents of the individual workbooks. - You can use external references to create reports that contain only relevant data, not the entire contents of the individual workbooks.

Split complex data into separate workbooks – Because you can link to portions of the worksheets, or create summaries, you can use multiple workbooks to store smaller parts of the data, instead of having it all in one giant workbook.

Part 1: Link Multiple Worksheets

Part 2: Refreshing Data in Multiple Worksheets

Background / Scenario

The bicycle shop keeps sales data in separate worksheets, one for each year. They would like you to perform data analysis on their sales records. In this lab, you will create a consolidated spreadsheet containing external links to each year’s data.

Required Resources

- Microsoft 365 Excel on Office.com
- Microsoft 365 OneDrive

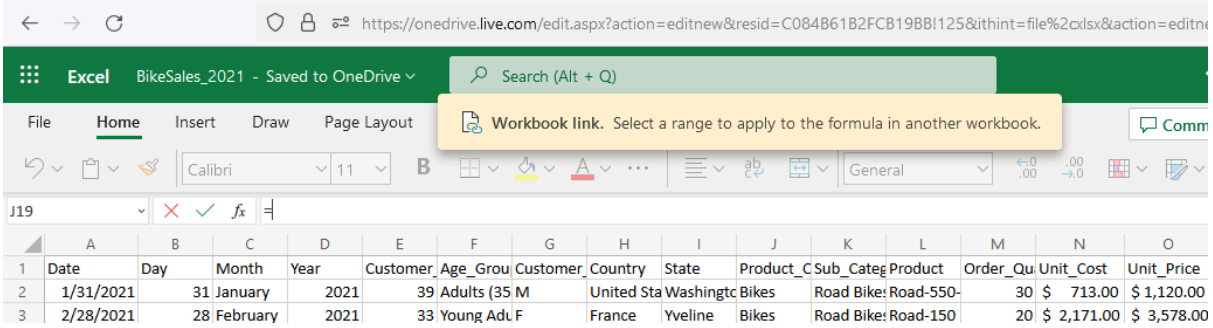
Instructions

Part 1: Creating External Links

Step 1: Open the worksheets to be linked

To work with the data from the 2021 and 2022 sales, download the Excel workbooks **BikeSales_2021.xlsx** and **BikeSales_2022.xlsx**.

a. Sign into your Microsoft 365 account. Upload the two workbook files to your Microsoft 365 OneDrive.
b. Open each of the workbooks in Excel online. You should see both workbooks listed on separate tabs in your browser.
c. On a separate browser tab, open a new blank workbook in Excel. Rename the spreadsheet **BikeSales_Consolidated**.
d. Place the cursor in the cell A1 of the **BikeSales_Consolidated** spreadsheet. **Type an equal sign (=)** and then click the browser tab containing the **BikeSales_2021** worksheet. A notice will appear that indicates you are setting a Workbook link.



e. Select the data that you want to link to the **BikeSales_Consolidated** workbook. In this case, select all of the data by placing the cursor in cell A1 and pressing the shortcut CTRL-SHIFT-END or by clicking and dragging from cell A1 to cell R14.
f. Return to the **BikeSales_Consolidated** workbook. The formula for the link appears in the formula bar and in cell A1. Press **ENTER**.
Note: If a warning appears that “External Data Connections have been disabled”, select Enable Content to continue.
g. Open a new worksheet in worksheet in the **BikeSales_Consolidated** workbook using the + tab at the bottom of the window. Repeat the previous steps in this new worksheet with the 2022 Sales worksheet.
h. The **BikeSales_Consolidated** workbook should now have two sheets, **Sheet1** containing the 2021 sales data and **Sheet2** containing the 2022 data.

Part 2: Refreshing Data in Multiple Sheets

The Worksheet Links function enables you to update data when the original data changes, by refreshing the linked data.

Step 1: Use Excel Functions to Summarize Data from Multiple Worksheets

a. Add a new worksheet (Sheet3) to the BikeSales_Consolidated workbook.
b. In **Row 1**, create two column headings: **2021 Average Cost** and **2021 Average Revenue**.
c. In Cell **A2**, type (do not copy and paste) **=AVERAGE(** Then, with the cursor next to the open parenthesis, click on the browser tab that contains the BikeSales_2021 worksheet.
d. Select the range of cells **Q2:Q14**. Return to the **BikeSales_Consolidated** workbook and press **ENTER**.

What is the value in Cell A2 after the spreadsheets are linked?

Answer Area

$26,479.30769

e. Place the cursor in cell B2. Repeat the steps to return the average of 2021 revenue (**Column R** in the **BikeSales_2021** Worksheet).

What is the value in Cell B2 after the spreadsheets are linked?

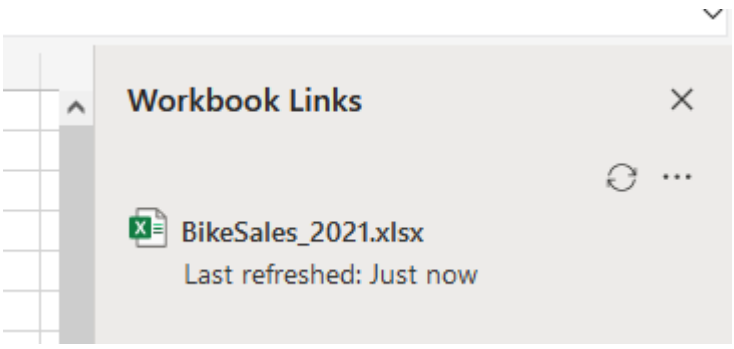
Answer Area

$43,764.31

Step 2: Update Data in the Source Spreadsheet

Now that the spreadsheets are linked, updates to the data in the source sheet can be automatically updated in the consolidated sheet.

a. Open the browser tab containing the **BikeSales_2021** worksheet.
b. Update the value in cell **Q6** to be $30,000.00. Return to the **BikeSales_Consolidated** worksheet **Sheet3**.
c. Select Data > Workbook Links (in the Queries & Connections section) in the tool bar. A window opens on the side of the screen with a list of workbooks linked to this worksheet.



d. Click the circular refresh icon next to the **BikeSales_2021.xlsx** listing.

What happens to the average values in cell A2?

Answer Area

$28,495.92

e. Open Sheet1 that contains the linked data from BikeSales_2021 and press the refresh button in the Workbook Links panel again.

What is the value in cell Q6?

Answer Area

$30,000

Reflection

What are some situations in which you would want to use Workbook links to keep data synchronized between workbooks?

This function in excel is very useful because it enables us to link workbooks in separate sheets and update it accordingly. This makes it convenient because we don’t have to cramp every one of our data and make it less confusing.

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